

A Disproof of Conjecture 00000001231

On the Jacobian order of the Johnson graph $J(n, k)$

TLMC Submission

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Abstract

Conjecture 00000001231 asserts that the order of the Jacobian (spanning-tree torsion group) of the Johnson graph $J(n, k)$ equals

$$|\text{Jac}(J(n, k))| = \binom{n-2}{k-1}^{\binom{n}{k}-1} \cdot \prod_i f(i), \quad f(i) = (i^2 - i + 1)^{e_i},$$

for some exponents e_i . We show that the stated formula is *false as written*: every factor $i^2 - i + 1$ is odd, so the product term is always odd, whereas at the admissible point $(n, k) = (4, 1)$ the true value of the Jacobian order is $\tau(K_4) = 16$, which is even. The matrix-tree theorem itself is used as the measuring device and is not disputed.

1 Setup

For a finite connected graph G , the *Jacobian* (a.k.a. sandpile group) of G is a finite abelian group whose order equals the number $\tau(G)$ of spanning trees of G . By the *matrix-tree theorem*,

$$\tau(G) = \det L^\circ, \tag{1}$$

where L is the graph Laplacian and L° is any $(|V| - 1) \times (|V| - 1)$ principal minor obtained by deleting one row and the corresponding column.

The *Johnson graph* $J(n, k)$ has as vertices the k -subsets of $\{1, \dots, n\}$, with two vertices adjacent exactly when their intersection has size $k - 1$ (equivalently, their symmetric difference has size 2).

Lemma 1. $J(n, 1)$ is the complete graph K_n .

Proof. Every 1-subset $\{x\}$ intersects every other 1-subset $\{y\}$, $x \neq y$, in the empty set, which has size $1 - 1 = 0 = k - 1$; hence all pairs of distinct vertices are adjacent, and there are $\binom{n}{2}$ edges. This is exactly K_n . $\square$

2 The counterexample $(n, k) = (4, 1)$

By Lemma 1, $J(4, 1) = K_4$. Its Laplacian and the relevant 3×3 principal minor are

$$L(K_4) = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}, \quad L^\circ = \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}.$$

Expanding $\det L^\circ$ along the first row:

$$\begin{aligned}\det L^\circ &= 3 \cdot \det \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} - (-1) \cdot \det \begin{pmatrix} -1 & -1 \\ -1 & 3 \end{pmatrix} + (-1) \cdot \det \begin{pmatrix} -1 & 3 \\ -1 & -1 \end{pmatrix} \\ &= 3(9 - 1) + 1(-3 - 1) - 1(1 + 3) \\ &= 24 - 4 - 4 \\ &= 16.\end{aligned}$$

This agrees with the Cayley–Prüfer value $\tau(K_n) = n^{n-2} = 4^2 = 16$. Hence

$$|\text{Jac}(J(4, 1))| = \tau(K_4) = 16. \quad (2)$$

3 The parity obstruction

Lemma 2 (Parity of the small factors). *For every integer i , the number $i^2 - i + 1$ is odd.*

Proof. $i^2 - i = i(i - 1)$ is a product of two consecutive integers, one of which is even; hence $i^2 - i$ is even and $i^2 - i + 1$ is odd. $\square$

Proposition 1 (The conjecture fails at $(4, 1)$). *There are no nonnegative integers e_i such that*

$$\binom{2}{0}^{\binom{4}{1}-1} \cdot \prod_i (i^2 - i + 1)^{e_i} = 16.$$

Proof. The leading factor is $\binom{2}{0}^3 = 1$, so the conjecture asserts $\prod_i (i^2 - i + 1)^{e_i} = 16$. By Lemma 2 each factor $i^2 - i + 1$ is odd, and any product of odd integers is odd. But 16 is even. Contradiction. $\square$

4 A second instance

$J(4, 2)$ is the line graph $L(K_4)$, i.e. the octahedral graph, with 6 vertices and 12 edges. Deleting the first row and column of its Laplacian and evaluating the resulting 5×5 determinant (carried out exactly in the accompanying `reproduce.py`) gives $\tau(J(4, 2)) = 384 = 2^7 \cdot 3$. The conjecture’s leading factor at $(4, 2)$ is $\binom{2}{1}^{\binom{4}{2}-1} = 2^5 = 32$, leaving a quotient $384/32 = 12$, which is again even — impossible for a product of odd factors $i^2 - i + 1$. The parity obstruction is thus not an artifact of a single point.

5 Scope of the disproof

Remark 1. We refute only the *literal* formula of Conjecture 00000001231, at the concrete admissible point $(n, k) = (4, 1)$ (with corroboration at $(4, 2)$). The matrix-tree theorem, invoked by the conjecture itself, is of course correct and is used here only as a tool. We make no claim about what corrected formula, if any, the author intended.

6 Machine verification

The directory `lean4/` contains a Lean 4 project (core only, toolchain `leanprover/lean4:v4.33.1`) that formally establishes, with zero axioms and no `sorry`:

- `Tlmc1231.tau_K4`: the 3×3 integer determinant above equals 16;
- `Tlmc1231.f_odd`: $\forall i : \mathbb{N}, (i \cdot i - i + 1) \bmod 2 = 1$;
- `Tlmc1231.prod_f4`: $\prod_{i < 4} (i^2 - i + 1) = 21$, together with `prod_f4_odd` and `sixteen_even`;
- `Tlmc1231.refute`: the parity-incompatible instance $\neg(\prod_{i < 4} (i^2 - i + 1) = 16 \wedge \tau = 16)$.

The script `reproduce.py` recomputes $\tau(K_4) = 16$, $\tau(J(4, 2)) = 384$, the leading factors 1 and 32, and the parity of $i^2 - i + 1$ for $0 \leq i \leq 100$.